使用RCF檔案設定Cisco FC交換器

ONTAP MetroCluster

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使用RCF檔案設定Cisco FC交換器

將Cisco FC交換器重設為原廠預設值

在安裝新的軟體版本和RCFs之前、您必須先清除Cisco交換器組態並執行基本組態。

您必須在MetroCluster 整個過程中、針對每個採用支援的FC交換器重複上述步驟。

> 顯示的輸出適用於Cisco IP交換器、但這些步驟也適用於Cisco FC交換器。

步驟

1. 將交換器重設為原廠預設值：

   a. 刪除現有的組態：+「寫入清除」

   b. 重新載入交換器軟體：+「重新載入」

      系統會重新開機並進入組態精靈。在開機期間、如果您收到「Abort Auto Provisioning」（中止自動資源配置）提示並繼續正常設定？（是/否）[n]、您應該回應「* yes *」以繼續進行。

   c. 在組態精靈中、輸入基本的交換器設定：

      - 管理密碼
      - 交換器名稱
      - 頻外管理組態
      - 預設閘道
      - SSH服務（遠端支援代理程式）完成組態精靈後、交換器會重新開機。

   d. 出現提示時、輸入使用者名稱和密碼以登入交換器。

      以下範例顯示登入交換器時的提示和系統回應。角括弧（「」<「」）會顯示您輸入資訊的位置。

```
---- System Admin Account Setup ----
Do you want to enforce secure password standard (yes/no) [y]:y
**<<<**

    Enter the password for "admin": password  **<<<**
  Confirm the password for "admin": password  **<<<**
         ---- Basic System Configuration Dialog VDC: 1 ----

This setup utility will guide you through the basic configuration of
the system. Setup configures only enough connectivity for management
of the system.

Please register Cisco Nexus3000 Family devices promptly with your
supplier. Failure to register may affect response times for initial
service calls. Nexus3000 devices must be registered to receive
entitled support services.

Press Enter at anytime to skip a dialog. Use ctrl-c at anytime
to skip the remaining dialogs.
```

e. 在下一組提示中輸入基本資訊、包括交換器名稱、管理位址和閘道、並輸入「* rSA*」作為SSH金鑰、如範例所示：

```
Would you like to enter the basic configuration dialog (yes/no): yes
  Create another login account (yes/no) [n]:
  Configure read-only SNMP community string (yes/no) [n]:
  Configure read-write SNMP community string (yes/no) [n]:
  Enter the switch name : switch-name **<<<**
  Continue with Out-of-band (mgmt0) management configuration?
(yes/no) [y]:
    Mgmt0 IPv4 address : management-IP-address  **<<<**
    Mgmt0 IPv4 netmask : management-IP-netmask  **<<<**
  Configure the default gateway? (yes/no) [y]: y **<<<**
    IPv4 address of the default gateway : gateway-IP-address  **<<<**
  Configure advanced IP options? (yes/no) [n]:
  Enable the telnet service? (yes/no) [n]:
  Enable the ssh service? (yes/no) [y]: y  **<<<**
    Type of ssh key you would like to generate (dsa/rsa) [rsa]: rsa
**<<<**
    Number of rsa key bits <1024-2048> [1024]:
  Configure the ntp server? (yes/no) [n]:
  Configure default interface layer (L3/L2) [L2]:
  Configure default switchport interface state (shut/noshut)
[noshut]: shut **<<<**
  Configure CoPP system profile (strict/moderate/lenient/dense)
[strict]:
```

最後一組提示會完成組態設定：

```
The following configuration will be applied:
  password strength-check
  switchname IP_switch_A_1
vrf context management
ip route 0.0.0.0/0 10.10.99.1
exit
  no feature telnet
  ssh key rsa 1024 force
  feature ssh
  system default switchport
  system default switchport shutdown
  copp profile strict
interface mgmt0
ip address 10.10.99.10 255.255.255.0
no shutdown

Would you like to edit the configuration? (yes/no) [n]:

Use this configuration and save it? (yes/no) [y]:
2017 Jun 13 21:24:43 A1 %$ VDC-1 %$ %COPP-2-COPP_POLICY: Control-Plane
is protected with policy copp-system-p-policy-strict.

[##########################################] 100%
Copy complete.

User Access Verification
IP_switch_A_1 login: admin
Password:
Cisco Nexus Operating System (NX-OS) Software
.
.
.
IP_switch_A_1#
```

2. 儲存組態：

```
IP_switch_A_1# copy running-config startup-config
```

3. 重新啟動交換器、然後等待交換器重新載入：

```
IP_switch_A_1# reload
```

4. 在其他三個MetroCluster 交換器上重複上述步驟、以進行「支援不整」的組態設定。

下載並安裝Cisco FC交換器NX-OS軟體

您必須將交換器作業系統檔案和RCF檔案下載到MetroCluster 位在整個過程中的每個交換器。

此工作需要檔案傳輸軟體、例如FTP、TFTP、SFTP或scp。 將檔案複製到交換器。

這些步驟必須重複執行MetroCluster 到位在整個過程中的每個FC交換器上。

您必須使用支援的交換器軟體版本。

"NetApp Hardware Universe"

> 顯示的輸出適用於Cisco IP交換器、但這些步驟也適用於Cisco FC交換器。

步驟

1. 下載支援的NX-OS軟體檔案。

   "Cisco下載頁面"

2. 將交換器軟體複製到交換器：

   「copy sftp：//root@server-ip-address/tftpboot/NX-os-name-name bootflash：VRF管理」

   在此範例中、「nxos.7.0.3.I4.6.bin」檔案會從SFTP伺服器10.99.99複製到本機bootflash：

```
IP_switch_A_1# copy sftp://root@10.10.99.99/tftpboot/nxos.7.0.3.I4.6.bin
bootflash: vrf management
root@10.10.99.99's password: password
sftp> progress
Progress meter enabled
sftp> get   /tftpboot/nxos.7.0.3.I4.6.bin
/bootflash/nxos.7.0.3.I4.6.bin
Fetching /tftpboot/nxos.7.0.3.I4.6.bin to /bootflash/nxos.7.0.3.I4.6.bin
/tftpboot/nxos.7.0.3.I4.6.bin                 100%  666MB   7.2MB/s
01:32
sftp> exit
Copy complete, now saving to disk (please wait)...
```

3. 在每個交換器上、確認交換器NX-OS檔案是否存在於每個交換器的bootflash目錄中：

   《Dir bootflash》

   以下範例顯示檔案位於「ip_switch_a_1」上：

```
IP_switch_A_1# dir bootflash:
                  .
                  .
                  .
  698629632    Jun 13 21:37:44 2017  nxos.7.0.3.I4.6.bin
                  .
                  .
                  .

Usage for bootflash://sup-local
 1779363840 bytes used
13238841344 bytes free
15018205184 bytes total
IP_switch_A_1#
```

4. 安裝交換器軟體：

   「install all system bootflash：nxos.version-number.bin kickstart bootflash：nxos.version：kickstart-number.bin」

```
IP_switch_A_1#  install all system bootflash:nxos.7.0.3.I4.6.bin
kickstart bootflash:nxos.7.0.3.I4.6.bin
Installer will perform compatibility check first. Please wait.

Verifying image bootflash:/nxos.7.0.3.I4.6.bin for boot variable
"kickstart".
[####################] 100% -- SUCCESS

Verifying image bootflash:/nxos.7.0.3.I4.6.bin for boot variable
"system".
[####################] 100% -- SUCCESS

Performing module support checks.
[####################] 100% -- SUCCESS

Verifying image type.
[####################] 100% -- SUCCESS

Extracting "system" version from image bootflash:/nxos.7.0.3.I4.6.bin.
[####################] 100% -- SUCCESS

Extracting "kickstart" version from image
bootflash:/nxos.7.0.3.I4.6.bin.
[####################] 100% -- SUCCESS
...
```

交換器軟體安裝完成後、交換器會自動重新開機。

5. 等待交換器重新載入、然後登入交換器。

   交換器重新開機後、會顯示登入提示：

```
User Access Verification
IP_switch_A_1 login: admin
Password:
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2017, Cisco and/or its affiliates.
All rights reserved.
.
.
.
MDP database restore in progress.
IP_switch_A_1#

The switch software is now installed.
```

6. 確認交換器軟體已安裝：

   《如何版本》

   下列範例顯示輸出：

```
IP_switch_A_1# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2017, Cisco and/or its affiliates.
All rights reserved.
.
.
.

Software
  BIOS: version 04.24
  NXOS: version 7.0(3)I4(6)   **<<< switch software version**
  BIOS compile time:  04/21/2016
  NXOS image file is: bootflash:///nxos.7.0.3.I4.6.bin
  NXOS compile time:  3/9/2017 22:00:00 [03/10/2017 07:05:18]


Hardware
  cisco Nexus 3132QV Chassis
  Intel(R) Core(TM) i3- CPU @ 2.50GHz with 16401416 kB of memory.
  Processor Board ID FOC20123GPS

  Device name: A1
  bootflash:   14900224 kB
  usb1:               0 kB (expansion flash)

Kernel uptime is 0 day(s), 0 hour(s), 1 minute(s), 49 second(s)

Last reset at 403451 usecs after  Mon Jun 10 21:43:52 2017

  Reason: Reset due to upgrade
  System version: 7.0(3)I4(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin
IP_switch_A_1#
```

7. 在剩下的三個FC交換器上重複上述步驟MetroCluster 、以進行支援。

下載並安裝Cisco FC RCF檔案

您必須將RCF檔案下載到MetroCluster 以「支援」為基礎的各個交換器組態中。

這項工作需要檔案傳輸軟體（例如FTP、Trivial File Transfer Protocol（TFTP）、SFTP或Secure Copy Protocol

（SCP））、才能將檔案複製到交換器。

這些步驟必須重複執行MetroCluster 到位在整個過程中的每個Cisco FC交換器上。

您必須使用支援的交換器軟體版本。

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共有四個RCF檔案、MetroCluster 其中四個交換器的RCF設定各一個。您必須針對所使用的交換器機型使用正確的RCF檔案。

交換器	RCF檔案
FC_SWIT_A_1	「NX323232_v1.80交換器-a1.txt」
FC_SWIT_A_2	「NX323232_v1.1_Switch-a2.txt'
FC_SWIT_B_1	「NX323232_v1.80交換器-B1.txt」
FC_SWIT_B_2	「NX323232_v1.1_Switch-b2.txt'

顯示的輸出適用於Cisco IP交換器、但這些步驟也適用於Cisco FC交換器。

步驟

1. 從下載Cisco FC RCF檔案 "RCF下載頁面MetroCluster"。
2. 將RCF檔案複製到交換器。
   a. 將RCF檔案複製到第一個交換器：

      「copy sftp：//_root@ft-server_ip-address/tftpboot/switch-specif_bootflash：VRF管理」

      在此範例中、「NX323232_v1.m_Switch-a1.txt' RCF檔案會從位於「10.10.99.99」的SFTP伺服器複製到本機bootflash。您必須使用TFTP/SFTP伺服器的IP位址、以及需要安裝的RCF檔案名稱。

```
IP_switch_A_1# copy sftp://root@10.10.99.99/tftpboot/NX3232_v1.8T-
X1_Switch-A1.txt bootflash: vrf management
root@10.10.99.99's password: password
sftp> progress
Progress meter enabled
sftp> get   /tftpboot/NX3232_v1.80_Switch-A1.txt
/bootflash/NX3232_v1.80_Switch-A1.txt
Fetching /tftpboot/NX3232_v1.80_Switch-A1.txt to
/bootflash/NX3232_v1.80_Switch-A1.txt
/tftpboot/NX3232_v1.80_Switch-A1.txt          100% 5141     5.0KB/s
00:00
sftp> exit
Copy complete, now saving to disk (please wait)...
IP_switch_A_1#
```

a. 對其他三個交換器重複上一個子步驟、請務必將相符的RCF檔案複製到對應的交換器。

3. 在每個交換器上、確認每個交換器的「bootflash」目錄中都有RCF檔案：

《Dir bootflash:》

下列範例顯示檔案存在於ip_switch_a_1上：

```
IP_switch_A_1# dir bootflash:
                  .
                  .
                  .
       5514    Jun 13 22:09:05 2017  NX3232_v1.80_Switch-A1.txt
                  .
                  .
                  .

Usage for bootflash://sup-local
 1779363840 bytes used
13238841344 bytes free
15018205184 bytes total
IP_switch_A_1#
```

4. 將相符的RCF檔案從本機bootFlash複製到每個交換器上的執行組態：

「copy bootflash：*switch-specific-RCF*.txt執行組態」

5. 將RCF檔案從執行中的組態複製到每個交換器的啟動組態：

「copy running-config startup-config」

您應該會看到類似下列的輸出：

```
IP_switch_A_1# copy bootflash:NX3232_v1.80_Switch-A1.txt running-config
IP_switch_A_1# copy running-config startup-config
```

6. 重新載入交換器：

   "重裝"

```
IP_switch_A_1# reload
```

7. 在其他三個MetroCluster 交換器上重複上述步驟、以進行REIP組態設定。

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